



# Assay

THE ASSAY OFFICE FOR AI

## A Protocol for Verifiable AI Model Evaluation

*Adaptive psychometrics on a tamper-evident ledger*

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**Abstract.** Public AI benchmarks have entered a trust crisis: leaderboards are gamed through selective reporting and vote manipulation, static test sets leak into training data, and the organisations that grade frontier models are increasingly funded by the labs they rank. I argue that the root cause is structural — reliability is **asserted**, never **measured** — and I make it measurable. I define a five-axis reliability scorecard (verifiability, calibration, contamination-resistance, manipulation-cost, statistical efficiency) and present **Assay**, a three-layer protocol that targets each axis by construction. A **measurement layer** replaces raw accuracy with adaptive Item-Response-Theory psychometrics; a **protocol layer** adds pre-registration, commit-reveal item cycles and model attestation; a **trust layer** notarises every evaluation on a public, tamper-evident ledger with identity-bound human voting. In a controlled proof of concept — including a red-team on 57,477 real LMArena votes — the design reaches the ranking quality of a 240-item static benchmark with 4× fewer items, neutralises a memorisation attack from +3.9 to 0.0 rank positions, and raises the cost of a two-position ranking manipulation from effectively zero to a bounded, identity-traceable expenditure. The limits of the approach are stated explicitly: predictive validity cannot be guaranteed a priori, and end-to-end cryptographic verifiability of closed models awaits practical zkML. A companion disclosure policy (§10) states precisely which components are open and which are held as operational secrets.

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**Author's note.** *This work did not begin as an academic exercise. I build production software systems for a living, and the question of which model to trust — and why the public numbers kept failing that question — turned into an engineering problem I could not route around. Assay is my attempt to hold the measurement of these systems to the same discipline we demand of the systems themselves: calibrated instruments, pre-registered procedures, and results that anyone can check without taking my word for them.* — N. F., Rovato, August 2026

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# 1 The trust crisis in AI evaluation

Benchmarks are the price signal of the AI economy. Procurement decisions worth billions, safety-critical deployment choices, and the allocation of research attention all pivot on a handful of leaderboard numbers. Yet the machinery that produces those numbers was built for a low-stakes academic world and is now buckling under adversarial pressure. Three failure modes are, by 2026, thoroughly documented.

**Contamination.** Public test sets are published on the web; frontier models are trained on the web; the questions and their answers therefore leak into training data. When researchers rewrote grade-school arithmetic problems in the style of GSM8K but with fresh numbers, several leading models dropped by up to double-digit percentage points [8] — they had memorised the exam, not learned the subject. A benchmark that cannot distinguish memorisation from capability is not measuring capability.

**Manipulation and selective reporting.** Because a leaderboard is an optimisation target, it is optimised. The 2025 study **The Leaderboard Illusion** documented how large providers could privately test many variants of a model on a popular arena and publish only the best score, while receiving a disproportionate share of the arena’s live data [9]. In the most-cited incident, a chat-tuned “arena special” variant of a frontier open model topped the board while a materially different model shipped to the public. Goodhart’s law [13], restated for the present setting: when a benchmark becomes a target, it ceases to be a good benchmark.

**Conflict of interest.** The organisations that grade models increasingly depend on the graded. A prominent “independent” mathematics benchmark was revealed to have been funded by a lab that also had access to most of its problems — a fact disclosed only after that lab announced a record on it. Independent leaderboard operators have raised nine-figure venture rounds and monetise the very labs they rank. An evaluator that depends financially on the parties it evaluates cannot credibly certify them.

The common denominator is not malice but **architecture**. In today’s benchmarks, reliability is a claim made by the scorer, unverifiable by anyone else, on a static artefact that decays the moment it is published. Assay’s thesis is that this is fixable, and that the fix is not a better test set but a different **contract** between evaluator, model provider, and the public.

## What this paper is, and is not

This is a **systems and methods** paper. Its contribution is the composition of known techniques — psychometrics, cryptographic commitments, mechanism design — into a protocol whose reliability is **measurable and independently checkable**. No single component is novel in isolation; the synthesis, and its application to AI evaluation, is. I publish the methodology in full because a service whose value proposition is verifiability cannot have a secret methodology. What I **do** hold back — item content, honeypot placement, and certain operational parameters — is stated explicitly in §10, and none of it is load-bearing for the scientific claims below.

# 2 Reframing reliability as a measurable quantity

The first contribution is definitional. “Reliability” is used in benchmark marketing as an adjective; I make it a vector of five quantities, each with an operational estimator and a status-quo baseline. This scorecard is the yardstick against which the rest of the protocol is judged — and, I propose, against which any evaluation service should be judged.

**A<sub>1</sub> Verifiability** share of published numbers a third party can reproduce  
 The share of published scores that an outside party can re-derive from committed artefacts, without trusting the scorer. For self-reported numbers today this is  $\approx 0\%$ . Assay targets **100% by construction**: every score is a deterministic function of a committed input bundle (§5–6).

**A<sub>2</sub> Calibration** empirical coverage of the declared  $(1-\alpha)$  interval  
 The empirical frequency with which a declared “ $\theta \pm \varepsilon$ ” interval contains the true value, against its nominal  $(1 - \alpha)$ . Most leaderboards publish point estimates with no interval at all. Assay publishes credible intervals and holds itself to nominal coverage (§4.2).

**A<sub>3</sub> Contamination resistance** rank distortion under partial test-set memorisation  
 The rank displacement obtainable by having seen the test in training. Under classical accuracy it is substantial (§8); Assay combines honeypots and fresh re-testing to drive it toward zero (§4.5).

**A<sub>4</sub> Manipulation cost** expenditure to move one's own rank by one position  
 The expenditure required to displace one's own rank. On an anonymous vote-based board it is effectively zero. Assay raises it to a bounded, **identity-traceable** cost and confines motion to within the declared uncertainty band (§7).

**A<sub>5</sub> Statistical efficiency** ranking quality per administered item  
 The ranking fidelity obtained per unit of expensive, leakable test content. Adaptive psychometrics extracts materially more than fixed-form accuracy (§8), which lowers cost and shrinks the contamination surface simultaneously.

There is a sixth axis I name but explicitly **exclude** from the guarantees. **Predictive validity** — whether a rank predicts real downstream utility — is the most important property of all and the one no methodology can guarantee a priori, precisely because of Goodhart's law. I treat it as measurable only **ex post**, and §9 states this limit plainly. A service honest about A<sub>6</sub> is, I would argue, more trustworthy than one that pretends to own it.

### 3 Architecture: three layers

Assay is organised as a stack (Figure 1). Each layer is responsible for a distinct class of threat, and each is independently substitutable. Reading top to bottom: the **measurement layer** decides what a score **is**; the **protocol layer** decides how a score is **produced and committed**; the **trust layer** decides how a score is **verified without a trusted operator**.

## The Assay stack

*Psychometrics measures · protocol prevents gaming · cryptography certifies*

### MEASUREMENT LAYER

#### *Adaptive psychometrics – the algorithm*

- 3PL Item Response Theory · Bayesian ability estimation (EAP)
- Computerised Adaptive Testing (Fisher-max, exposure control)
- Honeypot contamination detection · defended Bradley–Terry

### PROTOCOL LAYER

#### *Rules that make cheating structurally impossible*

- Pre-registration · commit–reveal item cycles
- Item-bank rotation · confidence-band rankings
- “What-you-test-is-what-you-ship” model attestation

### TRUST LAYER

#### *Verifiability without a trusted operator*

- Public tamper-evident ledger (L2 notarisation)
- Proof-of-personhood identity for human votes
- Confidential compute · zkML (roadmap)

### Verifiable, hallmarked evaluation

Each published number is reproducible from artefacts committed to the ledger.

*The novelty is the synthesis: no single component is new; the composition is.*

Figure 1: The Assay stack. Psychometrics measures, the protocol prevents gaming, cryptography certifies. Layers are substitutable: the measurement layer works today; the trust layer’s strongest component (zkML) is on the roadmap.

The remaining sections specify each layer, then give the threat model (§7), the empirical proof of concept (§8), the limitations (§9), and the disclosure and governance model that make the service credible (§10–11).

## 4 Measurement layer: adaptive psychometrics

Classical benchmarking reports the fraction of a fixed question set a model answers correctly. This is a maximum-likelihood estimate of a latent ability under the most restrictive possible model – one where every item is equally informative and equally difficult – and it discards almost everything psychometrics has learned since the 1950s. Assay’s measurement layer treats model evaluation as a testing problem and imports the corresponding machinery: Item Response Theory, Bayesian ability estimation, and adaptive item selection.

### 4.1 The item response model

Each item  $i$  carries three calibrated parameters: discrimination  $a_i > 0$ , difficulty  $b_i$ , and a pseudo-guessing floor  $c_i \in [0, 1)$  (for a  $k$ -way multiple-choice item,  $c_i \approx 1/k$ ). A model with scalar ability  $\theta$  answers item  $i$  correctly with probability given by the three-parameter logistic (3PL) model:

$$P_i(\theta) = c_i + \frac{1 - c_i}{1 + \exp(-a_i(\theta - b_i))}.$$

The 3PL is the workhorse of high-stakes human testing (the SAT, the GRE, medical licensing). Its virtue here is that it separates **what an item measures** (its  $a_i, b_i, c_i$ ) from **what a model is** (its  $\theta$ ), so that difficulty and capability live on the same scale and a score is no longer hostage to the particular

questions that happened to be on the test. Item parameters are themselves estimated — **calibrated** — from response data during a pre-registration phase; the calibration methodology and the resulting parameter tables are part of the proprietary item bank (§10), but the model above is fully public.

## 4.2 Bayesian ability estimation and honest intervals

Given a model's responses  $\mathbf{u} = (u_1, \dots, u_n)$  on administered items, the likelihood under local independence is

$$L(\mathbf{u} \mid \theta) = \prod_{i=1}^n P_i(\theta)^{u_i} (1 - P_i(\theta))^{1-u_i}.$$

With a weakly-informative prior  $\pi(\theta) = \mathcal{N}(0, \sigma_0^2)$ , the posterior is  $p(\theta \mid \mathbf{u}) \propto L(\mathbf{u} \mid \theta)\pi(\theta)$ , and I report the **expected a posteriori** (EAP) ability together with a posterior credible interval:

$$\hat{\theta}_{\text{EAP}} = \int \theta p(\theta \mid \mathbf{u}) d\theta, \quad \text{SE}(\hat{\theta}) = \sqrt{\int (\theta - \hat{\theta}_{\text{EAP}})^2 p(\theta \mid \mathbf{u}) d\theta}.$$

The interval is not decoration: publishing  $\hat{\theta} \pm 1.96 \text{SE}(\hat{\theta})$  and **holding it to nominal coverage** is how axis A<sub>2</sub> is satisfied, and — as §7 shows — it is also the cheapest structural defence against manipulation, because motion smaller than the band is by definition not a ranking change. In the proof of concept the empirical coverage of the nominal 95% interval is 94–95% across conditions.

## 4.3 Adaptive item selection

The information an item carries about ability is the Fisher information, which for the 3PL is

$$I_i(\theta) = a_i^2 \frac{Q_i(\theta)}{P_i(\theta)} \left( \frac{P_i(\theta) - c_i}{1 - c_i} \right)^2, \quad Q_i = 1 - P_i,$$

and the test information adds up over administered items,  $I(\theta) = \sum_i I_i(\theta)$ , with  $\text{SE}(\hat{\theta}) \approx 1/\sqrt{I(\hat{\theta})}$ .

**Computerised Adaptive Testing** (CAT) exploits this: rather than administer a fixed form, it selects each next item to maximise information at the current ability estimate,

$$i^* = \arg \max_{i \in \mathcal{A}} I_i(\hat{\theta}),$$

where  $\mathcal{A}$  is the set of not-yet-administered, currently-eligible items. Two constraints make this safe in an adversarial, repeated setting. First, **exposure control**: instead of always choosing the single most informative item, the selector samples uniformly among the top- $k$  (a randomesque rule), bounding how often any item is shown and thereby how quickly it can be learned by repeated probing. Second, **content balancing**: selection is constrained to respect a blueprint over domains so that ability is not estimated from an accidental monoculture of items. The precise  $k$ , the exposure ceilings, and the blueprint weights are operational parameters (§10); the selection principle is public.

The pay-off is axis A<sub>5</sub>. Because a well-chosen 60-item adaptive test concentrates its questions near each model's ability, it attains the ranking fidelity that fixed-form accuracy reaches only beyond 240 items (§8), a roughly fourfold reduction in the test content that must be produced, protected, and eventually retired as exposure accumulates.

## 4.4 Item calibration and scale maintenance

The parameters  $(a_i, b_i, c_i)$  are estimated, not asserted. New items enter the bank unscored, seeded into live administrations alongside operational items; their parameters are then fit by marginal maximum likelihood [15], integrating the latent ability out of the likelihood,

$$L(\mathbf{a}, \mathbf{b}, \mathbf{c}) = \prod_{j=1}^M \int \prod_{i \in C_j} P_i(\theta)^{u_{ij}} (1 - P_i(\theta))^{1-u_{ij}} \pi(\theta) d\theta,$$

maximised by an EM scheme over quadrature nodes, where  $C_j$  denotes the item set administered to model  $j$ . Because the bank rotates (§5.3), successive calibration cycles are tied to a common scale by **anchor-item equating**: a small overlap set links each cycle's scale to its predecessor's, and drift in anchor behaviour is itself monitored — anomalous anchor drift is a contamination signal in its own right. Item fit is re-audited in production, and items whose empirical response curves depart

from their calibrated parameters are quarantined pending review. Calibration sample sizes, anchor proportions and drift tolerances are operational parameters (§10).

Two remarks bound the scope of the model. First, administration stops when the posterior standard error reaches a preset precision target or an item ceiling, so every model is measured to comparable uncertainty rather than to a fixed item count. Second, the scalar-ability 3PL is the operating baseline, not a commitment: a multidimensional extension — separate ability axes for, say, knowledge, reasoning and code — is a planned refinement that changes the estimator, not the protocol guarantees.

#### 4.5 Contamination detection: honeypots and fresh probes

Efficiency shrinks the contamination surface but does not close it. Assay adds active detection built on a simple asymmetry: **a model that has memorised the public test reproduces its answer key, including the parts of the key that are deliberately wrong.**

A small, secret subset of public items are **honeypots**: their published answer key is intentionally incorrect. An honest model of ability  $\theta$  selects the planted (wrong) answer only by chance; a contaminated model that has memorised the leaked key reproduces the planted answer at a rate far above chance. Let  $H$  be the honeypot subset and  $X$  the number of planted keys a model reproduces on  $H$ . Under the honest null — a model answering from item content alone at its fitted ability  $\hat{\theta}$  —  $X$  has a chance expectation  $E_0$  and variance  $V_0$  computable from the fitted response model, and the detector standardises the observed excess,

$$z_{\text{hp}} = \frac{X - E_0}{\sqrt{V_0}},$$

so that reproduction of a leaked key, as opposed to genuine answering, surfaces as a right-tail anomaly. A second, independent signal compares performance on the public set with performance on a **fresh probe** — a difficulty-matched, never-published set drawn from the same bank. A contaminated model overperforms its own fresh-set ability on the public set; the standardised public–probe gap,  $z_{\text{gap}}$ , is referenced against the cohort of evaluated models so that benign drift between the two sets cancels. The two statistics are combined by a multiplicity-adjusted decision rule, and flagged models are re-scored on fresh items only. The chance-expectation model, the combination rule, the flag threshold, the honeypot fraction and their per-cycle re-tuning are specified in the private technical annex (§10); the reference implementation behind §8 uses conservative illustrative values, with which the combined detector catches 100% of heavy contamination at a 0.8% false-positive rate.

#### 4.6 Ranking from pairwise human preference

For open-ended quality, Assay retains pairwise human preference but hardens the statistics. Given a blind comparison in which model  $a$  beats model  $b$ , the Bradley–Terry model [1] sets  $\Pr(a \succ b) = \sigma(s_a - s_b)$  with  $\sigma$  the logistic function and  $s_m$  a latent strength; strengths are fit by regularised maximum likelihood over all battles, and reported on the familiar Elo scale,  $\text{Elo}_m = s_m \cdot 400 / \ln 10 + 1000$ . Uncertainty is quantified by a nonparametric bootstrap over battles (percentile intervals from  $10^3$  resamples), and it is the interval, not the point rating, that the leaderboard displays. Two further modifications matter. Battles are **weighted** by the trust of the identity that cast them (§6.2), so that a flood of fresh, one-sided, unverified votes cannot move a strength as far as an equal number of established, identity-bound votes. And rankings are reported as **confidence bands**, not integer places: two models whose bootstrap Elo intervals overlap are declared tied. §7 shows, on real data, how much these two changes cost an attacker.

### 5 Protocol layer: making cheating structurally impossible

Better statistics raise the price of some attacks; the protocol layer removes others by construction. Its four mechanisms are procedural, not algorithmic, and several of the proof of concept’s strongest results come from here.

## 5.1 Pre-registration

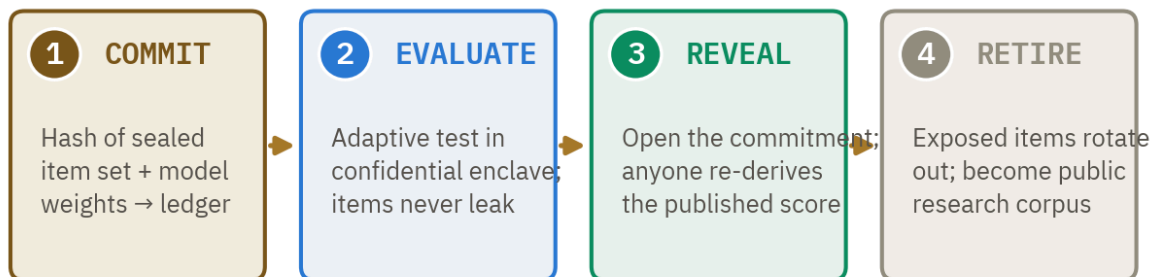
Every evaluation is **declared before it is run**: the model under test, its decoding parameters, the item blueprint, the metrics, and the analysis are committed in advance, and the result is published whatever it is. This single rule, borrowed from the clinical-trials world, eliminates the entire family of selective-reporting attacks — there is no “best of  $N$  runs” to cherry-pick when the run that counts is named in advance. In the proof of concept, publishing the best of twenty otherwise-identical runs buys a model +1.5 rank positions of pure noise; pre-registration sets that gain to zero by definition.

## 5.2 The commit–reveal lifecycle

Pre-registration is enforced cryptographically through a commit–reveal cycle (Figure 2). At **commit**, the scorer publishes a hash of the sealed item set and of the model artefact to the ledger, timestamping the evaluation without revealing its contents. At **evaluate**, the adaptive test is administered — for closed models, inside a confidential enclave (§6.3) so the items never leak. At **reveal**, the commitment is opened and any party can re-derive the published score from the committed bundle, satisfying verifiability ( $A_1$ ). At **retire**, exposed items rotate out of the active bank and are released as a public research corpus, so that today’s secret test becomes tomorrow’s open dataset.

Formally, the item commitment is a Merkle root [14],  $C = \text{root} \{H(s_i, x_i)\}_i$ , computed over the bank items  $x_i$  under per-item salts  $s_i$  with a cryptographic hash  $H$ . The construction is **hiding** — the root reveals nothing about item content — and **binding** — no committed item can be altered after the fact; at reveal, each item is published together with its Merkle inclusion proof, so a third party verifies membership without access to the rest of the bank. The pre-registered run manifest (Appendix C) is committed in the same manner, which is what makes the “no best-of- $N$ ” guarantee of §5.1 checkable rather than merely promised.

### Evaluation lifecycle: commit → evaluate → reveal → retire



*Priority is timestamped at COMMIT; integrity is public at REVEAL; contamination decays by design at RETIRE.*

Figure 2: The evaluation lifecycle. Priority is timestamped at commit; integrity is public at reveal; contamination decays by design at retire. The reveal step is what makes an Assay number checkable by someone who does not trust Assay.

## 5.3 Item-bank rotation

An item is a consumable, not a fixture. Because CAT concentrates exposure and honeypots detect leakage, Assay can measure the **effective half-life** of the bank and retire items on a schedule before their contamination signal degrades scores. This is the operational engine behind axis  $A_3$  over time, and it is also the item bank’s economic reality: the bank must be continuously replenished, which — as §11 notes — is the dominant recurring cost and the true competitive moat, since the **method** below is public and the **content** is not.

## 5.4 Model attestation: what-you-test-is-what-you-ship

The “arena special” attack — evaluating a variant that differs from the released model — is defeated by binding the score to a cryptographic identity of the artefact evaluated. For open models, the released weights are hashed and the certificate is valid only for that exact hash. For closed models reached

through an API, Assay pins a behavioural fingerprint at evaluation time and re-checks it, so that a silent swap between the evaluated and served model is detectable. A certificate therefore reads “**this hallmark applies to the model whose digest is 0x...**”, not to a name.

## 6 Trust layer: verifiability without a trusted operator

The trust layer answers the question the conflict-of-interest problem poses: why should anyone believe Assay, an organisation that could itself be captured? The answer is to remove Assay from the trusted computing base wherever possible.

### 6.1 Ledger notarisation

Commitments, reveals, model digests, and evaluation metadata are written to a public, tamper-evident ledger on a low-cost Layer-2 chain, where a write costs cents. I am precise about what this does and does not buy, consistent with the discipline of asking whether a blockchain is truly required: a centralised transparency log could provide append-only history, but a public chain adds the one property that matters here — **no party, including Assay, owns the register**. Because trust in an evaluator is exactly the thing that cannot be self-certified, and because the relevant distrust is genuinely multi-party (labs distrust the scorer and each other), this is one of the uncommon cases where public-ledger notarisation is the right tool rather than an affectation. It does one job — immutable, ownerless timestamping — and no more; the correctness of what is committed is a separate matter, handled above and below this layer.

### 6.2 Identity-bound voting

Human preference is only as trustworthy as the humans are real and distinct. Assay binds votes to **proof-of-personhood** identities and applies the trust-weighting and per-identity caps of §4.6. Pseudonymity, the default of naive on-chain designs, would make manipulation **worse**; the design goal is the opposite of anonymity. §7 quantifies, on real votes, how a modest verification cost per identity converts a free attack into an expensive and traceable one.

### 6.3 Confidential compute and zkML — roadmap, stated honestly

The hardest problem is evaluating a **closed** model verifiably: the provider will not release weights, yet a credible evaluation must prove that a specific model produced specific outputs on secret items. Two technologies address this, at different maturities. **Confidential computing** — hardware trusted execution environments with remote attestation — can today run a sealed model against sealed items so that neither leaks, and is the basis of Assay’s near-term closed-model path. **Zero-knowledge machine learning** (zkML) would go further, producing a cryptographic proof that “model with digest  $W$  produced output  $O$  on input  $I$ ” without revealing  $W$ ,  $O$ , or  $I$ ; recent work demonstrates zero-knowledge proofs for LLM-scale inference [12], but the cost is not yet practical at frontier scale. I therefore place zkML on the roadmap rather than on the critical path; the distinction is made explicit to avoid implying a guarantee the technology cannot yet support. The day zkML scales, end-to-end verifiable evaluation of closed models becomes possible; until then, confidential computing plus attestation is the honest state of the art.

## 7 Threat model and security analysis

I consider three adversaries: a **model provider** seeking to inflate its own rank; a **coordinated crowd** (bots or paid voters) manipulating preference; and — importantly — a **captured or negligent operator**, i.e. Assay itself acting in bad faith. For each I state the attack, the defending mechanism, and the residual risk. The point of the trust layer is that the third adversary is partially defended **against the operator’s own service**.

| Attack              | Vector                                  | Defence                                          | Residual risk                                       |
|---------------------|-----------------------------------------|--------------------------------------------------|-----------------------------------------------------|
| Contamination       | Train on leaked public test items       | Honeypots + fresh re-test + rotation (§4.5, 5.3) | Mild ( $\leq 10\%$ ) leakage detected less reliably |
| Selective reporting | Publish best of many private runs       | Pre-registration; commit-reveal (§5.1–5.2)       | Requires enforced commit discipline                 |
| Arena-special swap  | Evaluate a model $\neq$ the one shipped | Weight/API attestation (§5.4)                    | Closed-model swap needs behavioural pinning         |
| Ballot stuffing     | Inject fake pairwise votes              | Proof-of-personhood + weight + cap (§4.6, 6.2)   | Bounded, priced, traceable — not impossible         |
| Metric gaming       | Optimise to the published metric        | Confidence bands; rotating blueprint             | Goodhart effects mitigated, not eliminated          |
| Operator capture    | Scorer favours a paying lab             | Ledger reveal + governance firewall (§6.1, 11)   | Item content still trusts the operator              |

The manipulation analysis (A<sub>4</sub>) deserves quantification because it is the axis most often hand-waved. I ran a ballot-stuffing red-team against a Bradley–Terry ranking reconstructed from 57,477 real, publicly released LMArena votes over 64 models, targeting a mid-table model (rank 30, bootstrap rank interval 24–35). Figure 3 reports the illicit Elo gain as a function of injected votes, with and without Assay’s defences.

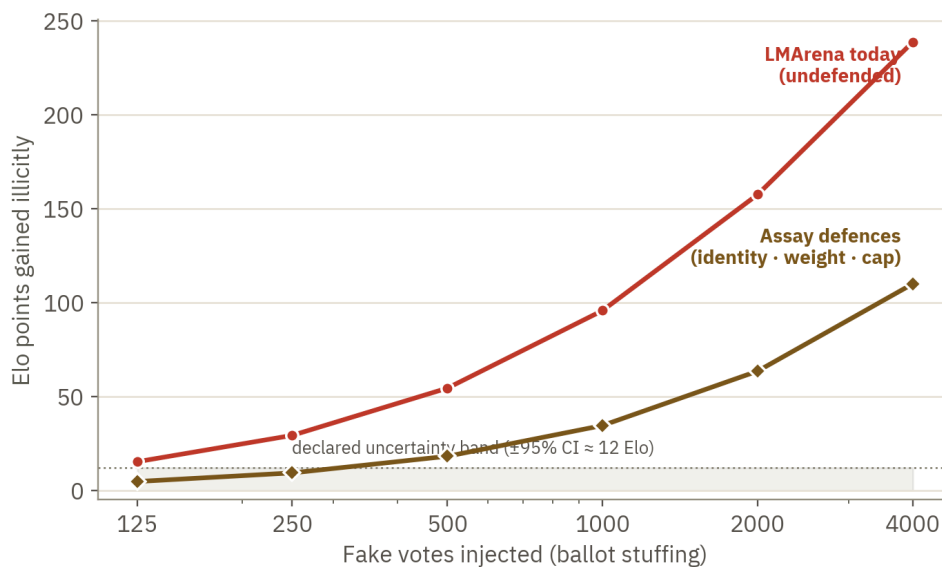


Figure 3: Ballot-stuffing on real LMArena data. Undefended (red), 41 fake votes — 0.07% of the corpus — buy two rank positions at zero cost. With Assay’s defences (brass), moving the target beyond its own declared uncertainty band requires 687 verified identities ( $\approx \text{€}414$ ) and leaves a permanent on-chain trace.

These defences do not make manipulation impossible, and no such claim is made here; assertions of manipulation-proof evaluation should themselves invite suspicion. What the defences achieve is a change of regime — a free, anonymous, untraceable attack becomes a **bounded, priced, and attributable** one — and the confidence-band convention renders sub-band motion meaningless by construction. Against the operator-capture adversary, the ledger’s reveal step means a captured scorer cannot silently alter a past number; what it cannot defend is the **content** of the items, which is why governance (§11), not cryptography, is the ultimate check on that last adversary.

## 8 Empirical proof of concept

I built the measurement engine from scratch and stress-tested the design in controlled simulation, plus one experiment on real data. The engine implements the 3PL model, EAP estimation on a

quadrature grid, CAT with exposure control, the honeypot and gap detectors, and defended Bradley–Terry. All results below are reproducible from the released code (Appendix B); the two simulated experiments use synthetic models of **known** ability, which is the only way to measure estimation error against ground truth, and is standard practice in psychometrics. The arena experiment uses real human votes.

### 8.1 Efficiency and calibration

Across 200 replications over 20 models, I compared three scorers at equal item budgets: classical accuracy (the status quo), IRT-EAP on the same fixed items (an ablation isolating the statistics), and adaptive CAT. Figure 4 summarises the comparison: adaptive CAT at 60 items reaches Kendall  $\tau = 0.948$  against ground-truth ranking, matching or exceeding classical accuracy at 240 items ( $\tau = 0.936$ ) — the fourfold efficiency of axis  $A_5$ . The honesty of the intervals ( $A_2$ ) holds too: empirical coverage of the nominal 95% credible interval is 94.6% at 60 items. I am careful not to over-claim: CAT-60 and IRT-EAP-240 are statistically comparable, so most of the gain is the better statistics and the last increment is adaptivity — the “4×” is measured against the **status quo that is actually deployed**, classical accuracy.

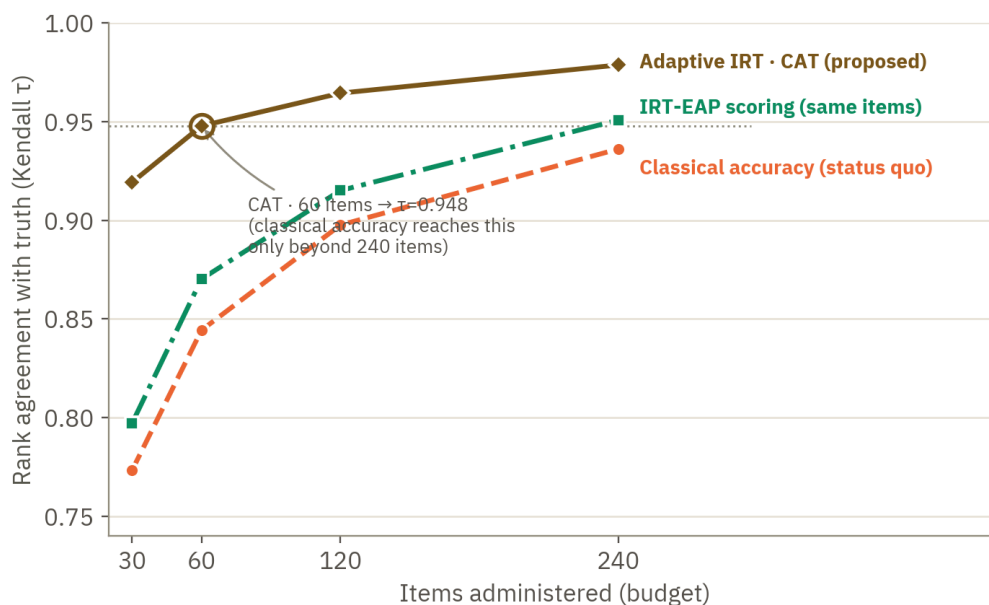


Figure 4: Ranking fidelity vs. item budget, 200 replications. Adaptive psychometrics reaches the fixed-benchmark quality of 240 items with roughly 60 — less content to produce, secure, and eventually burn.

### 8.2 Contamination

Six of 20 models were given memorised access to a fraction (10–50%) of the public set. Under classical accuracy a model that memorises half the test climbs +3.9 rank positions; with honeypots and fresh re-testing the same model returns to +0.0. Detection of heavy contamination is 100% at a 0.8% false-positive rate on honest models; mild (10%) contamination is caught 21% of the time — a real and stated limitation. On the full field, rank agreement with truth rises from  $\tau = 0.859$  (contaminated status quo) to  $\tau = 0.948$  after correction, essentially the contamination-free value ( $\tau = 0.941$ ). The defences do not merely reduce the problem; they return the ranking to almost exactly the clean-world case.

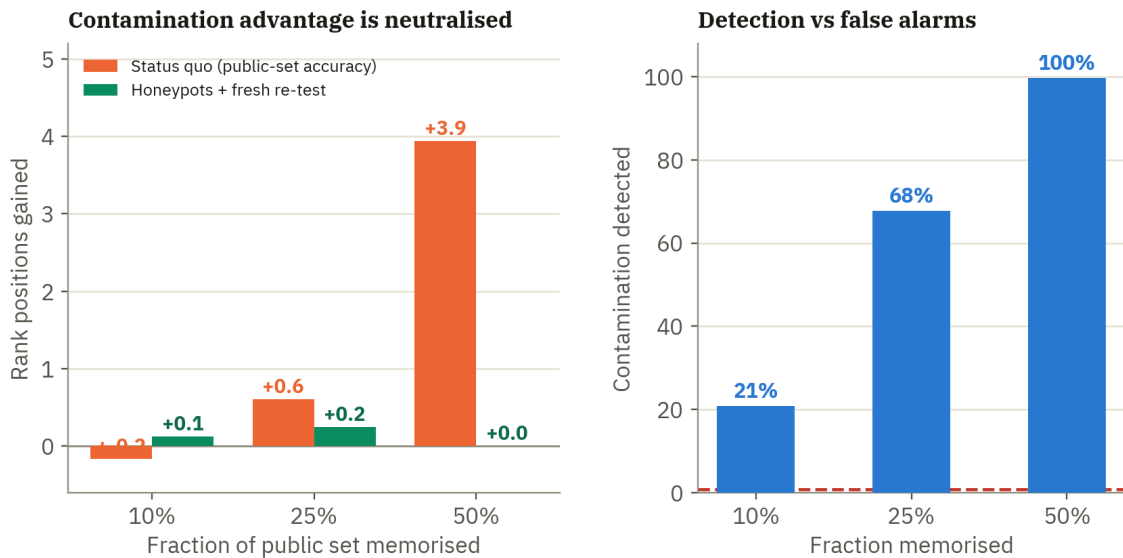


Figure 5: Left: a model memorising half the test gains +3.9 positions under the status quo, 0.0 under Assay. Right: detection rises with the intensity of cheating while false alarms on honest models stay under 1%. 300 replications.

### 8.3 Robustness

Every headline was repeated across five independent random seeds. The conclusions hold without exception: contamination correction beats the status quo in 5/5 seeds, interval coverage stays in the 93–97% band in 5/5, false positives stay below 2% in 5/5, and heavy-contamination detection stays at 100%. The numbers in this paper are not a favourable-seed artefact.

## 9 What Assay does not claim

The credibility of an evaluator is measured by what it refuses to assert. I therefore state the limits as first-class content, not fine print.

**Predictive validity is not guaranteed.** That a rank predicts downstream utility ( $A_6$ ) is the most important property and the one no methodology can secure a priori; Goodhart’s law erodes any published metric. Assay measures it only ex post and says so.

**The simulated experiments use synthetic models.** They demonstrate algorithmic properties against a known ground truth; they are not a substitute for validation on real, multi-dimensional models, which is Phase-1 work.

**Mild contamination remains hard.** Honeypots bite hardest against heavy memorisation; a model that has seen only a little is detected less reliably.

**Confidential compute and zkML are assumed, not implemented,** in the proof of concept. The secure execution of closed models is the hardest open engineering problem here.

**The algorithm is not the moat.** It is publishable and therefore copyable. The defensible assets are the accumulated item bank, the reputation, and the institutional discipline — capital and time, not code.

## 10 Disclosure policy and intellectual property

This section is itself part of the design, because it resolves the apparent contradiction between a **very detailed** public methodology and the protection of what is valuable. The resolution is the standard one from cryptography and anti-abuse systems: **publish the mechanism, keep the secrets that are meant to be secret.** The existence and design of the honeypot mechanism are public (above); the specific items that are honeypots are not, exactly as the existence of a spam filter is public while its live parameters are not.

**Published (this paper and the code)**

- The five-axis reliability scorecard and its estimators.
- The measurement mathematics: 3PL, EAP, Fisher-information CAT, MML calibration and anchor equating, defended Bradley-Terry.
- The honeypot and fresh-probe mechanisms, at concept level.
- The protocol: pre-registration, commit-reveal, rotation, attestation.
- The trust architecture and threat model.
- The proof-of-concept code and seeds (Appendix B).

**Withheld (private technical annex)**

- The item bank: content, calibrated parameters, difficulty tables.
- Which items are honeypots, and the planted-answer keys.
- Detector internals: chance-expectation models, statistic combination rules, flag thresholds.
- Operational parameters: exposure ceilings  $k$ , honeypot fractions, rotation schedule, identity-weight schedules.
- The item-authoring playbook and calibration pipeline.
- Fine-grained commercial parameters.

Three protections follow from, rather than contradict, publication. **Priority** is established by time-stamping this document's hash on a public ledger at release — using Assay's own method, which doubles as a demonstration. **Brand** is protected by trademark, the asset to which user trust actually attaches (see the accompanying brand and feasibility brief). **Defensive publication** is a feature, not a bug: a public method cannot be patented out from under the project, and in Europe mathematical methods and software “as such” are in any case weak patent ground, so I do not rely on patents. The realistic risk is not that a competitor clones the paper — they would still need the item bank, the reputation, and the capital — but that an incumbent adopts its best ideas; if so, the standard will have been set here, with dated priority, and the contest moves to the assets that compound. **This is a strategy statement, not legal advice; trademark and IP steps require qualified counsel.**

## 11 Governance and neutrality

Because the product **is** trust, the corporate form is load-bearing. Assay is structured as a **dual entity**, on the model of major open-standards foundations: a non-profit **foundation** owns the standard, the methodology, the item bank, and the ledger register — the assets whose neutrality is the value — while an operating **benefit company** builds tooling and sells buyer-side reports and can raise capital. The foundation controls the company, never the reverse. Three rules are written into the constitution: identical, published fees for all graded parties, with no premium services sold to those being ranked; no single funder or customer above a fixed share of revenue; and public accounts with external methodology audits. The revenue architecture deliberately leans on the **buyers** of models — enterprises and public bodies performing due diligence — rather than the **producers** being graded, to keep the thermometer independent of the patient. There is a structural irony I accept: the more authoritative the benchmark becomes, the larger a target it is, and the faster its bank must rotate — a permanent arms race, budgeted from day one.

## 12 Roadmap

**Phase 0 (validation, 6–9 months).** Re-run the proof of concept on real LLMs via API; author a pilot item bank with domain experts; publish this paper with a notarised hash; prototype the commit-reveal register on a Layer-2 chain. The goal is scientific credibility, not product. **Phase 1 (public MVP, 12–18 months).** A live adaptive leaderboard for a first domain, with the confidential-compute path for one closed model, and the first buyer-side certifications. **Phase 2 (standard, ongoing).** Multi-domain coverage, an accredited certification mark, and zkML pilots as the technology matures. Positioning as the independent European evaluation authority aligns with regulatory demand for third-party evaluation of frontier models.

## 13 Related work

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Assay stands on four literatures. From **psychometrics**: Item Response Theory and adaptive testing [2, 3], recently imported to LLM evaluation to cut sample sizes [10] and to put error bars on evals [11]. From **benchmarking**: the static suites whose saturation and contamination motivate this work — MMLU [4], GPQA [5], SWE-bench [6] — and the documented failures of leaderboards [9] and static sets [8]. From **preference modelling**: the Bradley–Terry model [1] underlying arena rankings [7]. From **verifiable computation**: zero-knowledge proofs for ML inference [12] and on-chain attestation. Assay’s contribution is not a new result in any of these; it is their composition into an evaluation protocol whose reliability is measurable and independently checkable — a gap none of the above fills alone.

## 14 Conclusion

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The trust crisis in AI evaluation is architectural, and architecture can be redesigned. By making reliability a measured vector rather than an asserted adjective, and by combining adaptive psychometrics, a commit–reveal protocol, and public notarisation, Assay targets four of five reliability axes by construction and is honest that the fifth — predictive validity — belongs to no one. The proof of concept, including a red-team on real data, shows the central claims are not aspirational: fourfold efficiency, contamination advantage returned to zero, and manipulation converted from free to priced and traceable. The remaining barriers are not primarily technical but economic and institutional — the item bank, the governance, the reputation. These are, in my view, the appropriate barriers for an institution whose product is trust; constructing them is the programme of work that follows from this paper.

## Appendix A — Notation

|                                 |                                                    |
|---------------------------------|----------------------------------------------------|
| $\theta$                        | latent model ability (log-odds scale)              |
| $a_i, b_i, c_i$                 | item discrimination, difficulty, pseudo-guessing   |
| $P_i(\theta)$                   | 3PL probability of a correct response to item $i$  |
| $I_i(\theta)$                   | Fisher information of item $i$ at ability $\theta$ |
| $\hat{\theta}_{\text{EAP}}$     | expected-a-posteriori ability estimate             |
| $z_{\text{hp}}, z_{\text{gap}}$ | honeypot and public–fresh gap detection statistics |
| $s_m$                           | Bradley–Terry latent strength of model $m$         |
| $\tau$                          | Kendall rank-correlation with ground truth         |

## Appendix B — Reproducibility

The proof-of-concept engine (`engine.py`) and experiments (`exp_static.py`, `exp_arena.py`, `robustness.py`) are released with master seed 20260818. The arena red-team uses the public dataset `lmsys/lmsys-arena-human-preference-55k` (57,477 battles, 64 models). Figures are generated with an accessibility-validated palette. All numerical claims in §8 are emitted directly by that code; this paper’s own SHA-256 digest is notarised at publication per §6.1.

## Appendix C — Pre-registration manifest (field summary)

Field semantics are public; the concrete values of each live cycle are published at reveal.

|                            |                                                                                      |
|----------------------------|--------------------------------------------------------------------------------------|
| <code>model_ref</code>     | digest of the released weights, or provider attestation identifier for closed models |
| <code>harness_ref</code>   | commit hash of the evaluation harness and scoring code                               |
| <code>blueprint_ref</code> | identifier of the domain blueprint constraining item selection                       |
| <code>decoding</code>      | declared decoding parameters: temperature, token limits, tool policy                 |
| <code>metrics</code>       | declared endpoints and scoring procedures                                            |
| <code>analysis_ref</code>  | identifier of the pre-registered statistical analysis                                |
| <code>commit_root</code>   | Merkle root of the sealed item set (§5.2)                                            |
| <code>registered_at</code> | ledger transaction anchoring this manifest before execution                          |

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